

Linear transformations between vector spaces

Definition

Suppose that V and W are vector spaces and $T: V \rightarrow W$ is a function from V to W . T is called a linear transformation from V to W if for all $u, v \in V$ and $c \in \mathbb{R}$,

$$T(u + v) = T(u) + T(v) \text{ and } T(cu) = cT(u).$$

Examples: A an $m \times n$ matrix then

$T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$. T_A is a lin. trans. in this sense. All linear trans. from $\mathbb{R}^n$ to $\mathbb{R}^m$ are of this form.

Ex: V = the v.sp. of polynomials in one variable x

Define D: $V \rightarrow V$ where

$$D(p) = \frac{dp}{dx}$$

$$\begin{aligned} \text{Check: } D(p+q) &= \frac{d(p+q)}{dx} = \frac{dp}{dx} + \frac{dq}{dx} \\ &= D(p) + D(q) \end{aligned}$$

$$D(cp) = \frac{d(cp)}{dx} = c \frac{dp}{dx} = c D(p).$$

The kernel and the range

Suppose that $T : V \rightarrow W$ is a linear transformation. Then we define

$$\underline{\ker}(T) = \{v \in V : T(v) = 0\} \text{ and } \underline{\text{rng}}(T) = \{T(v) : v \in V\}.$$

Ex: If $T = \underline{T}_A$ for some matrix A then

$$\ker(T) = \ker(\underline{T}_A) = \{v : \underline{T}_A(v) = 0\}.$$

$$= \{v : Av = 0\}.$$

$$= \text{Nul}(A), \text{ the nullspace of } A.$$

Ex: $D: \text{polys} \rightarrow \text{polys}.$

$$p \mapsto \frac{dp}{dx}$$

What is $\ker(D)$?

$\ker(D) = \underline{\text{all constant polys.}}$

is. the solutions of $D(p) = 0$ or $\underline{\frac{dp}{dx} = 0}$



are the constants.

differential
equation.

Ex: $\text{rng}(T_A)$ for A an $m \times n$ matrix?

$$T_A(v) = Av \rightarrow \in \mathbb{R}^m$$

$\uparrow$ in the column space of A .

$$\underbrace{\begin{pmatrix} | & \dots & | \\ u_1 & \dots & u_n \\ | & \dots & | \end{pmatrix}}_{\in \mathbb{R}^m} v \text{ is a lin. comb. of } u_1, \dots, u_n$$

$$\begin{aligned} \text{rng}(T_A) &= \text{column space of } A \\ &= \text{Col}(A) \end{aligned}$$

Ex: If we consider $D: V \rightarrow V$ given by diff.
what is the $\text{rng}(D)$?

$$\text{rng}(D) = V$$

Check: $p(x) = a_0 + a_1x + \dots + a_nx^n$

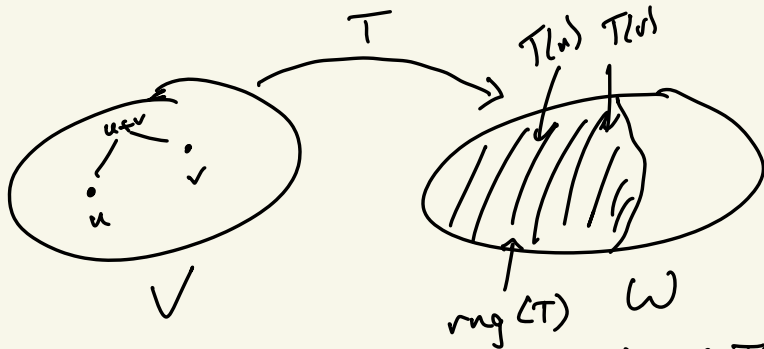
What q gives us $D(q) = p$?

$$q(x) = a_0x + \frac{a_1x^2}{2} + \frac{a_2x^3}{3} + \dots + \frac{a_nx^{n+1}}{n+1}$$

Claim: If $T: V \rightarrow W$ is a lin. trans. then
 $\ker(T)$ and $\text{rng}(T)$ are subspaces of
 V and W respectively.

Pf/ The case of $\ker(T)$ is very much like
the proof the nullspace is a subspace.

Why is the $\text{rng}(T)$ a subspace?



Choose two things in the range - $T(u)$ and $T(v)$
 for some $u, v \in V$. Why is $T(u) + T(v)$ also in
 the range? $T(u) + T(v) = T(u+v) \in \text{rng}(T)$.
 Scalars are similar!

Linear independence

If V is a vector space we say that $v_1, \dots, v_n \in V$ are linearly independent if whenever

$$c_1 v_1 + \dots + c_n v_n = 0$$

then $c_1, \dots, c_n$ are necessarily 0. A set of vectors B in V is linearly independent if every finite subset of B is linearly independent.

Comments! This notion agrees with lin. indep. in $\mathbb{R}^n$.

Ex: Claim: In P_n , ^{poly. of deg. $\leq n$} $1, x, x^2, x^3, \dots, x^n$ are
lin. indep.

$$\underbrace{c_0 \cdot 1 + c_1 x + c_2 x^2 + \dots + c_n x^n}_{\text{poly. of deg. } \leq n} = \underbrace{0}_{\text{zero function.}} \text{ in } P_n.$$

All c_i must be 0!

A basis for a vector space

If $\mathcal{B}$ is a subset of a vector space V then $\mathcal{B}$ is a basis for V if

- 1 $\mathcal{B}$ is linearly independent, and
- 2 $\mathcal{B}$ spans V .

Spanning set theorem

If V is a vector space and $H = \text{span}\{v_1, \dots, v_n\}$ then there is a subset of $\{v_1, \dots, v_n\}$ which is a basis for H .

Basis for the columnspace

The pivot columns of a matrix A form a basis for the columnspace for A . That is, the columns in the rref of A with leading 1's correspond to the columns of A which form a basis for the columnspace of A .